

"Glucometer"

Next-generation wearable devices allow patients with diabetes mellitus to continuously and non-invasively monitor their blood sugar levels. The values from individual devices are sent daily to a data center that compiles a single file containing the data of all patients connected to the monitoring service. The goal is to develop a program capable of extracting some statistics. Specifically, it should check for exceedances of the maximum level of 200 mg/dL.

Technical Specifications:

The program must handle an input file, `glucometers.txt`, containing data from an entire day of measurements. This file should be considered an aggregate of collected data and therefore contains, in random order, data for all patients.

The data for a single patient are in chronological order and may include missing measurements (intervals when the patient is not wearing the device). All measurements are taken at a regular frequency: one sample every 5 minutes.

Each row of the file contains 5 different data points separated by a single space, in the following order:

1. Patient identifier code consisting of 4 hexadecimal characters (format `PPPP`);
2. Acquisition time (format `hh:mm`);
3. Blood sugar level (g);
4. Body temperature (°C);
5. Heart rate (bpm).

The program, after storing the data in an appropriate data structure, must display a list of all patients who recorded at least one exceedance and, for each exceedance, the time and corresponding blood sugar level. The list must appear in order of severity, starting with the patient with the most exceedances.

Example Input File:

```
1BF0 17:00 160 37.0 68
1BF0 17:05 168 37.0 68
1BF0 21:00 180 37.3 66
1BF0 21:05 210 37.1 67
0AE1 21:10 187 37.3 69
0AE1 21:15 192 37.3 70
0AE1 21:20 195 37.4 70
0AE1 21:25 201 37.4 75
BBB3 22:30 108 37.5 73
BBB3 22:35 200 37.5 73
0AE1 23:05 203 37.4 73
0AE1 23:10 210 37.5 71
1BF0 21:10 213 37.2 68
```

Example Output:

```
0AE1 21:25 201
0AE1 23:05 203
0AE1 23:10 210

1BF0 21:05 210
1BF0 21:10 213

BBB3 22:35 200
```